Schedule and Agenda

- Today: PS in this room
  - Review case exam 1

Average	65,82609
Min	40
Max	81

  - Solve questions from the practice/study questions

- Now:
  - Simple linear regression: class exercise
  - Correlation
  - Multiple linear regression

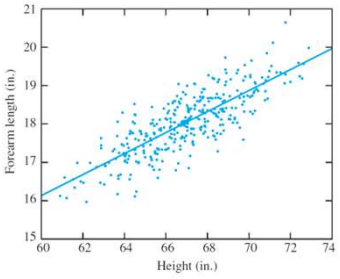
Correlation, r

When examining a scatterplot of bivariate data, we look at the,

- direction of the relationship (positive or negative),
- strength of the relationship,

and then we find a line that best fits the data.

Note: In computing correlation, we can only use quantitative data.



Computing *r*

- Let $(x_1, y_1), \ldots, (x_n, y_n)$ represent *n* points on a scatterplot.
- Compute the sample means and standard deviations of the *x*'s and *y*'s.
- Then convert each *x* and *y* to standard units. That is, compute the standardized scores:

$$\frac{(x_i - \bar{x})}{s_x} \qquad \frac{(y_i - \bar{y})}{s_y}$$

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

Correlation (1)

Population

$$Cov(X,Y) = E((X - \mu_X)(Y - \mu_Y))$$

$$Corr(X,Y) = \rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

population correlation coefficient

Sample

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

$$r_{xy} = r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$$

sample correlation coefficient

Computational Formula

- This formula is easier for calculations by hand:

$$r = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\sqrt{\sum_{i=1}^{n} y_i^2 - n\bar{y}^2}}$$

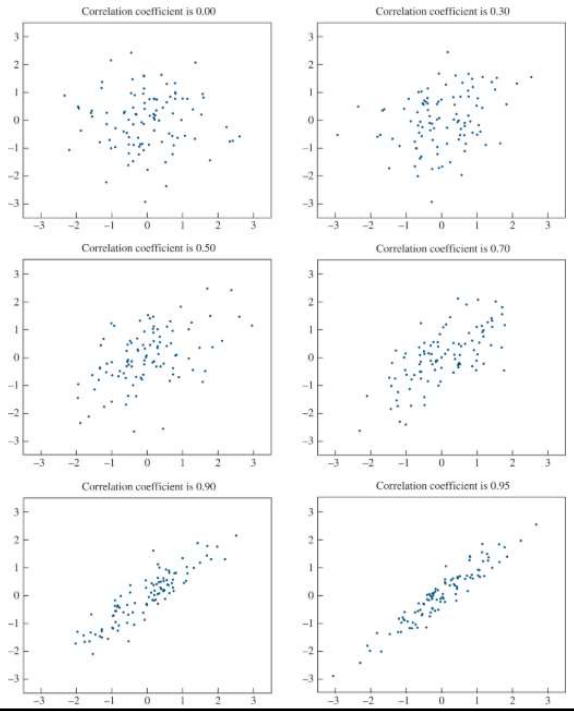
Correlation (2)

Properties	**Properties**
$Corr(aX+b, cY+d)$ $= Corr(X,Y)$ $= Corr(Y,X)$	$r_{ax+b,cy+d} = r_{x,y} = r_{y,x}$ *r: unitless*
$-1 \le \rho_{XY} \le 1$	$-1 \le r_{xy} \le 1$
$\rho_{xy} = 1$ or $\rho_{xy} = -1$ if $Y = aX + b$.	$r_{xy} = 1$ or $r_{xy} = -1$ if $y_i = ax_i + b$.
$\rho_{xy} = 0$ if X, Y are independent.	$(r_{xy})^2 = (r)^2 = r^2$ coefficient of determination

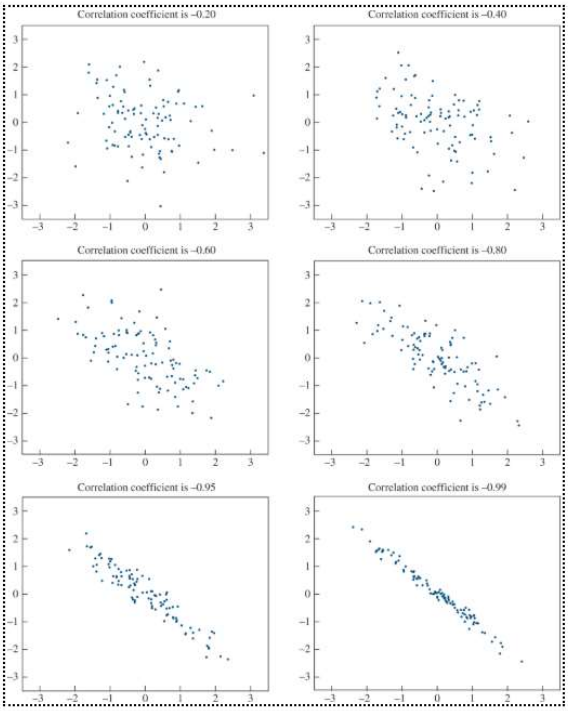
r Measures Only Linear Association

- The correlation coefficient should only be used when the relationship between the *x* and *y* is linear.
- Otherwise the results can be misleading.
- For example, two variables that are related quadratically will have a correlation coefficient of zero—they are certainly related, but not linearly.

Positive correlation



Negative correlation



9

Only linear relationship

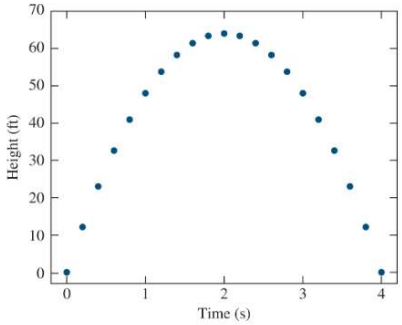


FIGURE 7.7 The relationship between the height of a free-falling object with a positive initial velocity and the time in free fall is quadratic. The correlation is equal to 0.

10

More Comments 2

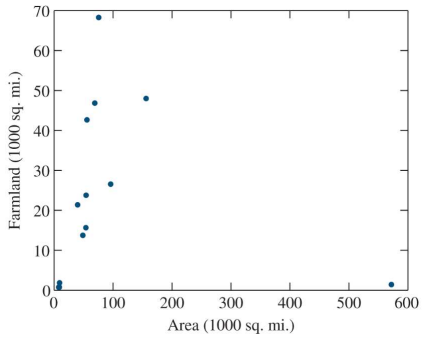
- If $r \neq 0$, then x and y are said to be **correlated**. If $r = 0$, then x and y are **uncorrelated**.

r Can Be Misleading when Outliers Are Present

- The correlation coefficient is often misleading for data sets that contain outliers.
- Outliers are a serious problem, as they make data more difficult to analyze.
- Some outliers are caused by data recording errors, or by failure to follow experimental protocol. These outliers can appropriately be corrected or deleted.
- Sometimes people delete outliers from a plot without cause, to give it a more pleasing appearance. This is not appropriate, as it results in an underestimation of the variability of the process that generated the data.

Example: effect of outliers

Figure 7.8 presents a plot of the area of farmland versus the total land area for a selection of U.S. states. In general, states with larger land areas have more farmland. The major exception is Alaska, which has a huge area but very little farmland. The point in the lower right corner of the plot, which represents Alaska, is an outlier, because it is detached from the main body of the data. The correlation for this scatterplot is $r = -0.12$, which indicates a weak negative relationship; in other words, it suggests that states with greater total area actually tend to have less farm area. But it is clear that there is a strong positive relationship, as one would expect, among the other states.



Correlation Is Not Causation

For children, vocabulary size is strongly correlated with shoe size. However, learning new words does not cause feet to grow, nor do growing feet cause one's vocabulary to increase.

There is a third factor, namely age, that is correlated with both shoe size and vocabulary.

- Older children tend to have both larger shoe sizes and larger vocabularies, and this results in a positive correlation between vocabulary and shoe size.

This phenomenon is known as **confounding**.

Confounding occurs when there is a third variable that is correlated with both of the variables of interest, resulting in a correlation between them.

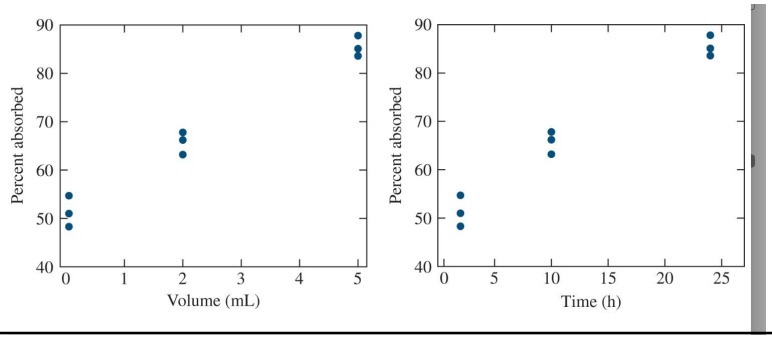
Example 1

Dependent variable: Rate of absorption of a certain chemical into skin.

Independent variable: volumes of the chemical on different pieces of skin ($r = 0.988$)

Independent variable 2: lengths of time that skin remain in contact with the chemical ($r = 0.987$)

She concludes that increasing the time that the skin is in contact with the chemical causes the percentage absorbed to increase as well. Are these conclusions justified?



15

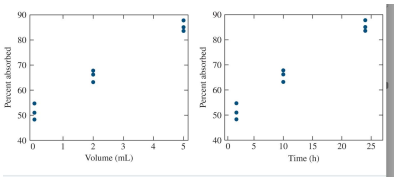
Example 1

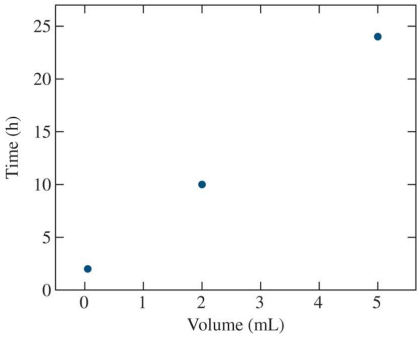
Dependent variable: Rate of absorption of a certain chemical into skin.

Independent variable: volumes of the chemical on different pieces of skin ($r = 0.988$)

Independent variable 2: lengths of time that skin remain in contact with the chemical ($r = 0.987$)

She concludes that increasing the time that the skin is in contact with the chemical causes the percentage absorbed to increase as well. Are these conclusions justified?





16

Correlation Is Not Causation

- Simply because two variables are correlated with each other, we cannot assume that a change in one will tend to cause a change in the other.
- Before we can conclude that two variables have a causal relationship, we must rule out the possibility of confounding.

Inference on the Population Correlation

- If the random variables *X* and *Y* have a certain joint distribution called a **bivariate normal distribution**, then the sample correlation *r* can be used to construct confidence intervals and perform hypothesis tests on the population correlation, ρ.

Inference on the Population Correlation

- Let $(x_1, y_1), \ldots, (x_n, y_n)$ be a random sample from the joint distribution of *X* and *Y* and let *r* be the sample correlation of the *n* points. Then the quantity,

$$W = \frac{1}{2}\ln\frac{1+r}{1-r}$$

is approximately normal with mean,

$$\mu_W = \frac{1}{2}\ln\frac{1+\rho}{1-\rho} \qquad \rho = \frac{e^{2\mu_W} - 1}{e^{2\mu_W} + 1}$$

and variance,

$$\sigma_W^2 = \frac{1}{n-3}.$$

Example 2

In a study of reaction times, the time to respond to a visual stimulus (*x*) and the time to respond to an auditory stimulus (*y*) were recorded for each of 10 subjects. Times were measured in ms.

x	161	203	235	176	201	188	228	211	191	178
y	159	206	241	163	197	193	209	189	169	201

Find a 95% confidence interval for the correlation between the two reaction times.

Solution

$$W = \frac{1}{2}\ln\frac{1+r}{1-r} = \frac{1}{2}\ln\frac{1+0.8159}{1-0.8159} = 1.1444$$

$$\sigma = \sqrt{1/(10-3)} = 0.3780$$

$$1.1444 - 1.96(0.3780) < \mu_W < 1.1444 + 1.96(0.3780)$$
$$0.4036 < \mu_W < 1.8852$$

$$\frac{e^{2(0.4036)}-1}{e^{2(0.4036)}+1} < \frac{e^{2\mu_W}-1}{e^{2\mu_W}+1} < \frac{e^{2(1.8852)}-1}{e^{2(1.8852)}+1}$$
$$0.383 < \rho < 0.955$$

21

Example 2

$H_0: \rho \leq 0.3$

$H_1: \rho > 0.3.$

Find the *P*-value

Under H_0 we take $\rho = 0.3$, so, using Equation (7.5),

$$\mu_W = \frac{1}{2}\ln\frac{1+0.3}{1-0.3} = 0.3095$$

The standard deviation of W is $\sigma = \sqrt{1/(10-3)} = 0.3780$. It follows that under H_0, $W \sim N(0.3095,\ 0.3780^2)$. The observed value of W is $W = 1.1444$. The z-score is therefore

$$z = \frac{1.1444 - 0.3095}{0.3780} = 2.21$$

The P-value is 0.0136. We conclude that $\rho > 0.3$.

22